

Complements of Operational Research

8.3 Multi-criteria decision making: sensitivity analysis

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The weights are opinions

In 8.1 the weights came out of a meeting: cost 0.482, range 0.272, payload 0.158, service 0.088.

In another meeting, with other people, other numbers would come out.

The right question is not “what is the true weight of cost?”, but

how far can the weight change without the decision changing?

If the answer is “a lot”, the decision is robust and the discussion of the weights can stop. If it is “a little”, that is where it is worth spending time.

The same idea as before

- ch. 1: Hurwicz with varying α ; subjective probabilities;
- ch. 2: simulating uncertain durations;
- ch. 7: the EOQ with its flat-bottomed optimum.

Two questions for the vans:

1. how heavy would cost have to be for C to overtake B?
2. and what if *all* the judgements had a small error?

Varying one weight

Fix the weight p of criterion k ; the others keep their proportions to each other and sum to $1 - p$:

$$w_j(p) = w_j \frac{1 - p}{1 - w_k}, \quad j \neq k.$$

Then the score of each alternative is a **straight line** in p :

$$S_i(p) = S_i(0) + p (p_{ik} - S_i(0)),$$

with $S_i(0)$ the score without criterion k and $S_i(1) = p_{ik}$ (only criterion k counts).

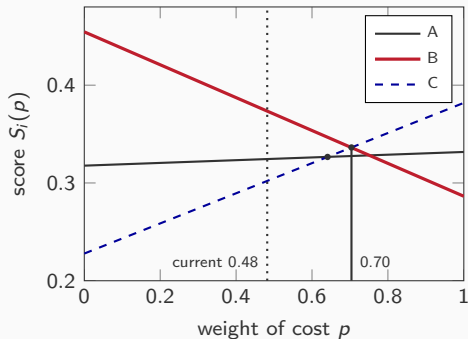
In the **Excel sheet on Moodle** the line goes through the current point (w_k, S_i) and through the point $(1, p_{ik})$:

$$S_i(p) = \frac{p - w_k}{1 - w_k} (p_{ik} - S_i) + S_i.$$

It is the same line.

Lines that cross inside $[0, 1]$ mark the weights at which the order of the alternatives changes.

The weight of cost



With the current weight (0.482), B leads comfortably.

- at $p = 0.641$, C overtakes A (swap in 2nd place);
- at $p = 0.704$, C overtakes B: **only here does the decision change.**

Cost would have to go from 48% to more than 70% of the total weight. And varying the weight of range, payload or service between 0 and 1, B always stays first.

Switching points

Two alternatives i and j tie when $S_i(p) = S_j(p)$.

With $d_0 = S_i(0) - S_j(0)$ and $d_1 = p_{ik} - p_{jk}$:

$$p^* = \frac{d_0}{d_0 - d_1} \quad (\text{if } 0 \leq p^* \leq 1).$$

B against C, cost criterion:

$S(0) = (0.318, 0.455, 0.228)$, so $d_0 = 0.2267$,

$d_1 = 0.2864 - 0.3819 = -0.0955$ and

$p^* = 0.2267/0.3222 = \mathbf{0.704}$.

Weights for which B stays first

crit	current weight	B first if
cost	0.482	$p \leq 0.704$
range	0.272	any p
payload	0.158	any p
service	0.088	any p

Reading: only cost can change the decision, and only if it weighs much more.

The table answers management's question without repeating the calculations for each weight.

What if the judgements were different?

One judgement: the cost–range comparison (it was 2).

a_{12}	w_{cost}	CR	S_A	S_B	S_C
1	0.416	0.013	0.324	0.384	0.292
2	0.482	0.005	0.324	0.373	0.302
5	0.556	0.067	0.325	0.362	0.313
9	0.589	0.158	0.326	0.356	0.318

B always wins; with 9 the matrix even stops being consistent. For C to tie with B, cost would have to be 7 times more important than range and 9 times more important than payload and service.

All the judgements: simulation with 20 000 replications; each of the 9 judgements (6 for the criteria, 3 for service) goes up one step, down one step or stays, at random, on the scale $1/9, \dots, 1/2, 1, 2, \dots, 9$.

Result

B comes first in **100%** of the replications (only 1.1% of the simulated matrices had $CR > 0.10$).

The decision does not depend on fine-tuning in the meeting — which is, after all, the best news an AHP can give.

When the margin is small

It is not always like this. In problems like those in the tests, the suppliers' scores are often less than 0.02 apart.

In those cases:

1. compute the switching points of the heaviest criteria;
2. if a switching point is close to the current weight, say so in the answer;
3. review the judgements on that criterion with the decision maker — or gather more information (firm prices, contractual lead times).

When answering an AHP problem

- the chosen alternative;
- the consistency (CR);
- the margin over the 2nd alternative;
- the criterion and the weight from which the choice changes.

A choice “by 0.01” with $CR = 0.09$ is not a firm recommendation: it is a technical tie.

```
import numpy as np

def pesos_com(w, k, p):          # weight p on k, rest proportional
    w = np.asarray(w, float); r = w.copy(); r[k] = 0
    r = r / r.sum() * (1 - p); r[k] = p
    return r

def viragem(M, w, k, i, j):      # weight of k at which i and j tie
    M = np.asarray(M, float); s0 = M @ pesos_com(w, k, 0)
    d0, d1 = s0[i] - s0[j], M[i, k] - M[j, k]
    p = d0 / (d0 - d1)
    return p if 0 <= p <= 1 else None

# M and w of the vans (8.2)
print(viragem(M, w, 0, 1, 2))    # 0.704: C overtakes B at cost weight 70%
```

sensibilidade.py also includes `sensibilidade(M, w, k)`, which returns the scores on a grid of weights (for the plot).

Try it: redo the simulation of the judgements with two steps instead of one. Does B still always win?

Example 6 — sensitivity in the supplier problem (2nd test, 27/06/2019)

Go back to Example 5 of 8.2.

- a) What is the margin between the chosen supplier and the second one?
- b) Up to what weight of the raw-material cost does the choice hold? Which supplier overtakes it?
- c) Below what weight of the transport cost does the choice change?
- d) What recommendation would you give the company?

Example 7 — the same for Example 3 (exam, 11/07/2019)

Repeat the analysis for the heaviest criterion. Is the decision more or less robust than in Example 6?

Exercise sheet 8 — Exercise 6 (homework)

Location of a health centre (new)

A town council is choosing between three sites, L1, L2 and L3, for a new health centre, with three criteria: population served within 15 minutes (L1 42 000, L2 35 000, L3 28 000 inhabitants), cost of the land and of the construction (3.2, 2.6 and 2.1 M€) and accessibility by public transport (L2 is moderately preferable to L1 (3), L1 is slightly preferable to L3 (2) and L2 is strongly preferable to L3 (5)). Population is moderately more important than cost (3) and slightly more important than accessibility (2); accessibility is slightly more important than cost (2).

- a) Criteria weights and consistency ratios.
- b) Local priorities and chosen site.
- c) The councillor for finance wants to give more weight to cost. From what weight of cost does the choice change? Which would be the new site?
- d) Below what weight of accessibility does the choice change? Comment on the robustness of the decision.

Submission: by email to the lecturer, by the deadline given in class. All exercises: **Exercise sheet 8**.

KEY TAKEAWAYS — 8.3

1. The weights come from judgements: the useful question is how far they can change without changing the decision.
2. Varying one weight (the others proportional), the scores are straight lines; the crossings give the switching points $p^* = d_0/(d_0 - d_1)$.
3. Perturbing all the judgements at random (simulation) shows whether the choice depends on fine-tuning.
4. Always answer with the choice, CR , the margin and the switching point of the most sensitive criterion.

End of chapter 8 and of the course.

The course in eight chapters

Chapter	The idea to keep
1 Decision theory	decision criteria and robustness of the choice
2 Project planning	models, uncertainty and simulation
3 Scheduling	exact and heuristic methods
4 Equipment replacement	marginal cost versus average cost
5 Dynamic programming	decomposition and the principle of optimality
6 Queueing	capacity, variability and service level
7 Inventory management	competing costs and a flat-bottomed optimum
8 Multi-criteria decision making (AHP)	judgements, consistency and robustness of the decision

After learning to optimise, the last question is always the same: **does the decision hold if the data or the judgements change?**

Model, optimise, compare and test the robustness of decisions.

References and resources

References Full list: *Course bibliography*.

- Hillier, F. S., Lieberman, G. J., *Introduction to Operations Research*, 11th ed., McGraw-Hill, 2021 — Decision Analysis.
- Triantaphyllou, E., Sánchez, A., A sensitivity analysis approach for some deterministic multi-criteria decision-making methods, *Decision Sciences*, 28(1), 151–194, 1997.
- Pacheco, M. C. R., Goldman, F. L., O AHP como um modelo matemático: uma análise de sensibilidade simples, 2016 (Moodle Example 1).
- von Adamovich, P. H. G., Wajnberg, V. B., *Aplicação do método AHP em um empreendimento da Eletrobras – Furnas*, Graduation project, UFRJ, 2016 (Moodle Example 3).

Resources

- Exercise sheet 8. `code/python/08_ahp/08_3_sensitivity/sensibilidade.py`. Moodle: Excel sheet “AHP — análise de sensibilidades” (AHP — sensitivity analysis).